

Complications

Physical and Psychosexual Implications

Anogenital warts are frequently perceived as disfiguring and can significantly impact sexual lifestyle. They may lead to feelings of anxiety, guilt, anger, and loss of self-esteem, and often raise concerns about future fertility and cancer risk.

Pre-cancer and Cancer

Premalignant lesions—such as vulval intraepithelial neoplasia (VIN), anal intraepithelial neoplasia (AIN), and penile intraepithelial neoplasia (PIN)—as well as invasive cancers (vulval, anal, and penile) can coexist with anogenital warts, develop within them, or be misdiagnosed as warts.

Bowenoid papulosis (BP) presents as reddish-brown lesions and is associated with oncogenic HPV types. It falls within the clinical spectrum of anogenital intraepithelial neoplasia. Clinical signs suggesting neoplastic transformation include bleeding, irregular pigmentation, ulceration, or lesions with palpable dermal infiltration. In such cases, urgent biopsy or specialist referral is warranted.

A rare variant of HPV 6/11 infection is the **giant condyloma acuminatum (Buschke-Löwenstein tumor)**, a form of verrucous carcinoma characterized by aggressive local invasion into surrounding tissues. Management requires specialist surgical and oncological input, with recent evidence indicating favorable outcomes using chemo-radiotherapy.

Diagnosis – Key Points

- Perform examination under adequate lighting; use a magnifying lens or colposcope for better visualization of small lesions.
- In men, always inspect the urethral meatus and maintain a low threshold for examining the perianal area and performing proctoscopy to evaluate the anal canal.
- In women, examine the perianal region and perform speculum examination to rule out cervical or vaginal lesions. Maintain a low threshold for proctoscopy to assess the anal canal. Proctoscopy is safe in individuals with perianal warts.
- Biopsy is not required for typical anogenital warts but must be performed if there is suspicion of precancer or cancer. It can also aid in differential diagnosis.
- Not all papular lesions are due to HPV. Consider normal anatomical variants such as **pearly penile papules** and **Fordyce spots**, as well as differential diagnoses including **molluscum contagiosum** and **seborrheic keratoses**.
- HPV typing of anogenital warts has no clinical utility and is not recommended.
- The acetic acid (vinegar) test may be used by some clinicians to detect subclinical HPV lesions, but its role in diagnosis and management remains controversial.

Management

These guidelines were informed by recent recommendations from national bodies in the United Kingdom and United States. Treatment evidence was evaluated using grading criteria from the Agency for Health Care Policy and Research (see Table 1).

Note that not all treatments for genital warts are available globally. This guideline focuses on therapies accessible in Europe, with clarification where availability is limited to specific countries.

Information, Explanation, and Advice for the Patient

- Patients should receive a clear, detailed explanation of their condition, supplemented with accurate written information.
- Reliable patient resources are available through the CDC (<http://www.cdc.gov/std/hpv>) and ASHA (<http://www.ashastd.org/std-sti/hpv.html>).

Therapy

Currently Available Treatments

A systematic review of randomized controlled trials (RCTs) on genital wart treatments was conducted (Woodhall SC, Lacey CJ, unpublished data), incorporating prior systematic reviews and primary trial data (see Search Strategy). Key findings are summarized in Table 2.

- Only **surgical therapies** achieve primary clearance rates approaching 100%.
- **Recurrence** occurs after all treatment modalities, with rates typically ranging from 20–30% or higher, including new lesions at previously treated or untreated sites.
- All treatments may cause **local adverse effects**, such as itching, burning, erosion, and pain.